

Extracting Accurate Long-Term Behavior Changes from a Large Pig Dataset

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Abstract: Visual observation of uncontrolled real-world behavior leads to noisy observations, complicated by occlusions, ambiguity, variable motion rates, detection and tracking errors, slow transitions between behaviors, etc. We show in this paper that reliable estimates of long-term trends can be extracted given enough data, even though estimates from individual frames may be noisy. We validate this concept using a new public dataset of approximately 20+ million daytime pig observations over 6 weeks of their main growth stage, and we provide annotations for various tasks including 5 individual behaviors. Our pipeline chains detection, tracking and behavior classification combining deep and shallow computer vision techniques. While individual detections may be noisy, we show that long-term behavior changes can still be extracted reliably, and we validate these results qualitatively on the full dataset. Eventually, starting from raw RGB video data we are able to both tell what pigs main daily activities are, and how these change through time.

1 INTRODUCTION

Table 1: A comparison of datasets on pigs

Paper	# Frames	# Annotated frames	Annotation types	# Pens	Acquisition time	# Pigs	Publicly available
(Seo et al., 2020),(Sa et al., 2019)	-	3,904	boxes	1	1 day	9	✗
(Brünger et al., 2020)	-	1,000	ellipses	2	4 months	-	✗
(Zhang et al., 2019)	-	22,200	boxes, IDs	1	3 days	9	✗
(Mittek et al., 2017)	2,100,000	-	pen boundaries, feeder, waterer	1	5 days	15	✗
(Cowton et al., 2019)	-	1,646	boxes, IDs	1	-	20	✗
(Li et al., 2019)	-	1,500	pigs' contours	1	7 days	4	✗
(Zhang et al., 2020; Li et al., 2020)	156,900	1000 (videos)	5 behaviors	3	80 days	9	✗
(Psota et al., 2019)	2,000	2,000	4 body parts locations	17	multiple weeks	variable	✓
(Psota et al., 2020)	135,000	135,000	3 body parts locations, IDs	9	multiple weeks	7-16	✓
Ours	3,429,000	7,200	boxes, IDs, 5 behaviors	1	23 days over 6 weeks	8	✓

Pork is the second most consumed meat (Transparency Research, 2019) across the world behind poultry, and more than 700 million (Shahbandeh, 2020) pigs were raised in 2019 alone.

Modern intensive pig farming is highly mechanized, with automation of the environmental temperature and airflow, supply of feed, water and the removal of wastes. Driven by efficiencies of scale, farms have also grown larger, and there has been a reduction in staff time per pig (Swan, 2020). As an example, in the EU more than half of the pork production comes from large intensive farms (Pol Marquer, 2020).

Behavior analysis could be used by farm staff, vets and scientists to reveal the pigs' state of health and welfare, but on most farms, a typical weaner-grower-finishing pig may only be briefly inspected once or twice a day as part of a large group. There is an increasing interest in using automated methods to monitor pigs' behavior on farm settings (Wurtz et al., 2019; Nasirahmadi et al., 2017). Aspects of behavior such as gait, use of different areas and resources in the pen, social clustering, activity can all be valuable information. Changes in behavior from the expected norm can be used as an early warning sign for behavior problems such as tail biting (D'Eath et al., 2018), social aggression (Chen et al., 2017), diseases (Fernández-Carrión et al., 2017), or for production issues such as thermal comfort (Costa et al., 2014). The use of cameras and various other sensor technologies in animal agriculture – to gather useful real-time data to guide management decisions – is often referred to as 'precision livestock farming' (Vranken and Berckmans, 2017).

In this work, we present a behavior analysis pipeline built on automatic pig detection and tracking, capable of providing a report of the changes in a set of 5 fundamental individual behaviors (lying, moving, eating, drinking and standing) through time.

While the same topics are of great interest in the scientific community when applied to humans ((Spinello and Arras, 2011; Stewart et al., 2016; Urtasun et al., 2006; Fleuret et al., 2007; Andriluka et al., 2008) among many others), less research exists addressing the same tasks in the animal domain. This may sound counter-intuitive at first, given that for some applications, like identification and tracking, working with animals completely avoids any privacy and security concerns. However, there is often a wide gap between the expertise of people working on the techniques (computer vision and machine learning scientists mainly) and those working directly with livestock (veterinary and biology researchers).

Recently, thanks to the democratization of computer vision and deep learning, numerous works have been presented for livestock and wildlife detection (Spampinato et al., 2008; Norouzzadeh et al., 2018; Seo et al., 2020; Sa et al., 2019; Psota et al., 2019), tracking (Underwood et al., 2013; Zhang et al., 2019; Mittek et al., 2017), identification (Bergamini et al., 2018; Liu et al., 2019a; Liu et al., 2019b) and also behavior analysis (Tschärke and Banhazi, 2016; Cowton et al., 2019; Li et al., 2019; Zhang et al., 2020; Li et al., 2020). Although techniques are now available, the increasing usage of deep convolutional neural networks has seen the demand for high quality annotated data soaring.

To this end, a contribution of this paper is also an unrestricted public pigs dataset, providing both manual and automatic annotation for multiple tasks, including detection, tracking, identification and behavior analysis.

In summary, the main contributions of our work are:

- A behavior analysis pipeline that focuses on individual pig behaviors to infer statistics about 5 different individual behaviors and how these change through time;
- Evidence that the behavior statistics at the aggregated week level are reliable and robust to error in the various steps of the pipeline;

- A public available dataset comprising 7200 fully annotated frames.

In the following, we present an overview of related works in Sec 2, with emphasis on public available swine solutions, and a description of the pig dataset in Sec 3. We then describe the detection-tracking-behavior pipeline in Sec 4 and we employ it to output statistical information about the pig behaviors over the full dataset that we discuss in Sec 5.

2 RELATED WORK

We present here an overview of the swine literature. We purposely decide not to categorize related works by their specific processes (e.g. tracking) because often tasks are approached in a sequential fashion by the same work (e.g. both detection and tracking can be performed for individual pig behavior analysis).

In (Seo et al., 2020) a TinyYOLO (Redmon et al., 2016) architecture is employed to detect pigs from infrared videos. Much focus is placed on execution speed, as the target platform is an embedded device. Images are acquired from a single pen and the training set includes 2904 images, while the test comprises 1000 images. The authors also approach the same task using traditional computer vision algorithms in (Sa et al., 2019). They propose a method to detect pigs under various illumination conditions by combining information from depth and infrared images, using spatio-temporal interpolation.

(Psota et al., 2019) take another approach and cast detection as a segmentation task. The targets are not bounding boxes anymore but instead 4 semantic parts of the animal (ears, shoulder and tail) which are detected using a Fully Convolutional Network. The Hungarian algorithm is then employed to link those parts for each individual pig. A dataset with 2000 images from multiple pens is publicly available online. The authors extend their work in (Psota et al., 2020), where they focus on tracking by leveraging the fixed cardinality of the targets. Their tracker achieves real-time performance and is based on features extracted from a CNN. Also the dataset of this work is publicly available.

Similarly, in (Brünger et al., 2020) the bounding boxes are replaced with ellipses, which are detected through a segmentation network. The intuition is that pigs are much closer to an ellipse in terms of shape when images are acquired from above. The dataset includes 1000 images recorded over a period of 20 days. 13 pigs from a single pen were recorded. An encoder-decoder architecture is trained with multiple losses to segment individual instances, using the notion of outer and inner edge of the animal.

In (Zhang et al., 2019) a Single Shot Detector (Liu et al., 2016) architecture is used to perform detection. A *tag-box* is then extracted from each detected animal to perform tracking using a variation of the MOSSE (Bolme et al., 2010) tracking algorithm. The dataset includes multiple pens and has been acquired over a period of 3 days. In total, 18000 images have been collected and annotated for the training set and 4200 for the test set.

(Mittek et al., 2017) leverage the depth signal from a Microsoft Kinect to fit 3D ellipses in an unsupervised fashion. The pen boundaries need to be annotated only once to define the working area of the following algorithm. Information from surface normals is employed to detect the boundaries between the pigs when these are very close. The dataset includes 2.1M frames from 5 consecutive days of a pen with 15 pigs.

In (Cowton et al., 2019) detection, tracking and behavior analysis of individual pigs is performed. First, R-CNN (Girshick et al., 2014) is used to detect bounding boxes, that are then input into two real-time tracker algorithms. Transfer learning is required to accommodate the covariate shift from a traditional deep learning dataset. Then, idle and moving behaviors are detected from tracklets. The dataset includes 1646 annotated images, which are split with 0.5 ratio between the training and test sets.

(Li et al., 2019) focus their attention on the mounting behavior only, which is identified as a cause of epidermal wounds and fractures. They collect a dataset from a week of acquisitions of a single pen with 4 young male pigs. 1500 frames are annotated with segmentation masks and mounting/no-mounting behavior flag. Then, a Mask R-CNN (He et al., 2017) is employed to detect and segment individual pigs. Finally, a multi-dimensional eigenvector is computed from the detected bounding-box and segmentation and classified into the two possible behaviors.

Differently, in (Zhang et al., 2020; Li et al., 2020) the behavior analysis is rephrased as an end-to-end video classification task. A dataset (PBVD-5) of 1000 short clips is collected and annotated with one out of five different behaviors (feeding, lying, walking, scratching and mounting), with 200 videos for each behavior. Data comes from 4 pens with up to 3 pigs in each. Then, in (Zhang et al., 2020) a two streams architecture employs both RGB and optical flow information to classify snippets and individual frames, and the results are fused using a

consensus function. The authors compare the performance of various architecture, including ResNet (He et al., 2016) and Inception (Szegedy et al., 2015) networks, as backbones.

To summarize, multiple works that tackle detection, tracking and behavior analysis of pigs exist. However, their focus is on the techniques only, while a thorough analysis of the application of those techniques on a wide dataset and the reliability of the computed statistics is still missing. In the following, we show that a combination of well established algorithms for the above mentioned tasks, even with their intrinsic limits due to the challenging setting, can be reliably employed to draw accurate long-term behavior changes statistics.

3 DATASET

The dataset was collected between 5 Nov and 11 Dec (2019, 6 weeks) in a single pigpen (5.8m x 1.9m) with 8 growing pigs at SRUC’s research pig unit (near Edinburgh, UK). The pigs were mixed intact males and females weighing around 30kg at the start of the study. They were given a 3-space feeder with ad libitum commercial pig feed, two nipple water drinkers and a plastic enrichment device (Porcichev, East Riding Farm Services Ltd, Yorkshire, UK) suspended at pig height from a chain. Pigs were also given straw and shredded paper on a part-slatted floor. Color image and depth data was collected using an Intel RealSense D435i camera positioned at 2.5 meters from the ground. Both RGB and depth information were acquired at 6fps with a resolution of 1280×720 , and the acquisition was limited to daytime (from 7AM to 7PM), due to the absence of artificial light during nighttime.



Figure 1: An example of depth and RGB data for the same frame. The depth data has several artifacts. One of the pigs in front of the feeder has a wide (black) spot with value zero, while one in the rear has both zero and out of distribution (white patch) areas.

The acquired frames were appended into video sequences of fixed size (1800 frames each corresponding to 5 minutes) for both compression efficiency and logical organization of the data. Figure 1 shows an example of RGB and depth information for the same frame. It is worth noting how the depth signal proved to be almost completely unreliable due to the presence of heavy non-white noise. Using it as an additional signal in our algorithms not only did not increase performance, but it even hinders it in several trials.

We acquired a total of 3,429,000 frames. Together with the raw data, we also provide manual annotations for different tasks for a subset (12 sequence corresponding to 7200 frames spread over the 6 weeks) of the dataset. These annotations were manually generated by 4 different people using a custom version of VaticJS (Bolkensteyn, 2016) available at <https://stefanopini.github.io/vatic.js/>. In each frame, the annotator:

- Draws a rectangular bounding box around each visible pig;
- Associates each bounding box with one of the 8 pigs using a numeric identifier;
- Selects a behavior among a list of 5 options (lie, move, eat, drink and stand).

The 12 sequences were annotated and split between training and validation to cover the entire time window of the acquisition process. This guarantees that the quality of the supervised algorithms employed in the rest of this work is representative of the full dataset.

Table 1 reports statistics for our dataset and compares it with others already published by the scientific community (both publicly and not). Although a bigger dataset (Psota et al., 2020) is publicly available, it only includes

3 key-points and IDs annotations. Contrarily, ours provides annotations for detection, tracking and behavior analysis. The dataset can be accessed at <https://aimagelab.ing.unimore.it/go/pigs-behaviours>.

4 BEHAVIOR ANALYSIS PIPELINE

Although the main focus of this work is understanding individual pig behaviors, several steps are required to fill the gap between raw data and behaviors. First, pigs need to be individually detected in each frame. The position information alone is already enough to identify behaviors that do not require temporal knowledge, such as eating or drinking. However, as other behaviors require multiple detections of the same pig in consecutive frames (e.g. moving or standing), we use tracking to associate the bounding boxes from consecutive frames into tracklets. A summary of the employed techniques for detection and tracking is given before focusing on behavior analysis. For both tasks we report supervised metrics on the annotated evaluation set.

Table 2: Metrics from the detector on the validation set. We report results for individual sequences as well as those from the whole validation set

Validation sequence	AP (%)	TP (%)	FP (%)	Missed (%)
A	84.63%	89.18%	10.82%	1.16%
B	97.28%	99.59%	0.41%	2.24%
C	100.00%	100.00%	0.00%	0.00%
D	95.75%	96.97%	3.03%	0.85%
E	98.38%	99.45%	0.55%	1.03%
Whole set	95.21%	97.04%	2.96%	1.06%

4.1 Detection

Pig detection is treated as a supervised computer vision task, powered by the ground truth annotations. A state-of-the-art deep convolutional neural network is used for multiple object detection, namely YOLO v3 (Redmon and Farhadi, 2018). We pre-train it on the ImageNet dataset (Deng et al., 2009) and fine tune it for pig detection by replacing the classification grid layer to predict only 2 classes (background and pig). Because the original dataset contains chiefly portrait pictures, we replaced the network anchors with a new set computed on our training set bounding boxes. Furthermore, since the camera depicts also parts of other pigs’ pens, we apply a mask on the video frames with the shape of the pen area containing the 8 pigs that we want to track. We set a threshold on the network’s confidence scores and we also apply non-maximum suppression using a threshold on the IoU between predicted boxes. We experimented with those two hyper-parameters but found that the default values (0.9 and 0.4 respectively) in practice worked well for the task. However, we include the a-priori knowledge of having a limited known number of entities we want to detect. As such, we always take up to 8 bounding boxes.

Table 2 shows results in terms of Average Precision (AP), number of true positives (TP), false positives (FP) and missed detections on the validation set. We report statistics for the individual sequences and the average on the full validation set.

The reported detection metrics are satisfactory. Figure 2 shows some detection failure cases from the validation set. Failures are mainly due to two reasons. First, differently from humans, pigs stay extremely close together most of the time, either while sleeping (usually on top of each other), fighting or just standing. In these conditions, it becomes extremely likely to have more than one pig per cell in the classification layer. Second,

Table 3: Metrics from the tracker on the validation set. We report results for individual sequences as well as those from the whole validation set

Validation sequence	MOTA (%)	IDF ₁ (%)	# Switches	# Fragmentations	# Tracklets	Avg. tracklet length (# frames)
A	76.78%	55.10%	23	187	24	597
B	97.35%	88.39%	12	13	17	834
C	100%	100.00%	0	0	8	1800
D	92.97%	88.46%	9	43	24	597
E	97.92%	78.29%	12	18	13	1104
Whole set	93.00%	82.05%	11.2	52.2	17.2	986.4

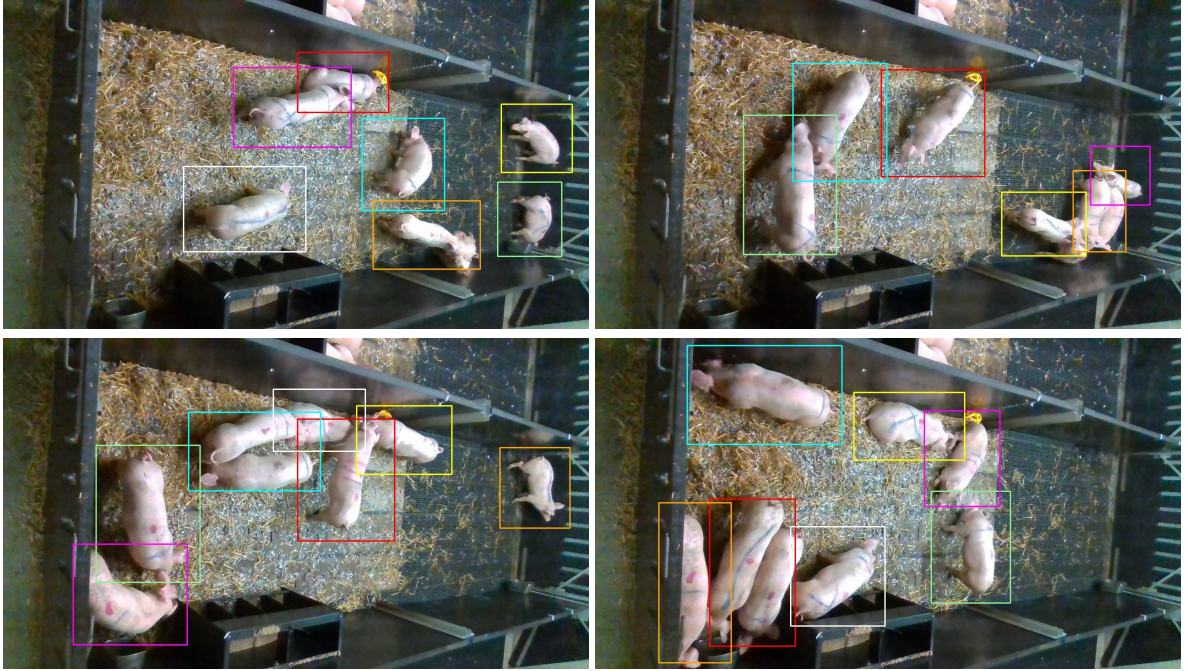


Figure 2: Examples of detection failures. A bounding box contains more than a single pig when the pigs are too close (e.g. red bounding box in bottom-right figure). Moreover, even when two separate bounding boxes are successfully generated for close pigs, they sometimes include portions of the other animal (e.g cyan bounding box in top-right figure)

bounding box annotations become less reliable when pigs cannot be contained individually by an axis-aligned rectangle.

These two factors pose a great challenge to algorithms designed for detecting humans. While other works use better fitting annotations that partially solve these issues (like ellipses in (Brünger et al., 2020)), these require custom algorithms to be handled and are more expensive to annotate compared to bounding boxes.

Although detection of single instances may be noisy, the amount of available data greatly reduces the noise influence. For example, we report in Fig. 3 the detected bounding box area on the full unlabeled dataset averaged by day and pig. It can be observed how the area increases monotonically (by around 45% throughout the entire acquisition window) which is expected when ad libitum food is available and only reduced activity can be performed due to space constraints. The 45% correlates well with the predicted increase in observed area of 67% based on $(W_{after} = 65\text{kgs}/W_{before} = 30\text{kgs})^{2/3}$, assuming that weight is proportional to volume.

4.2 Tracking

For tracking the pigs, we employ a simple yet effective tracking-by-detection algorithm (Pini et al., 2019) that groups into tracklets consecutive detections of the same pig. In practice, for each new detection a new tracker is created and initialized. In the following frames, updated trackers and single-frame detections are matched

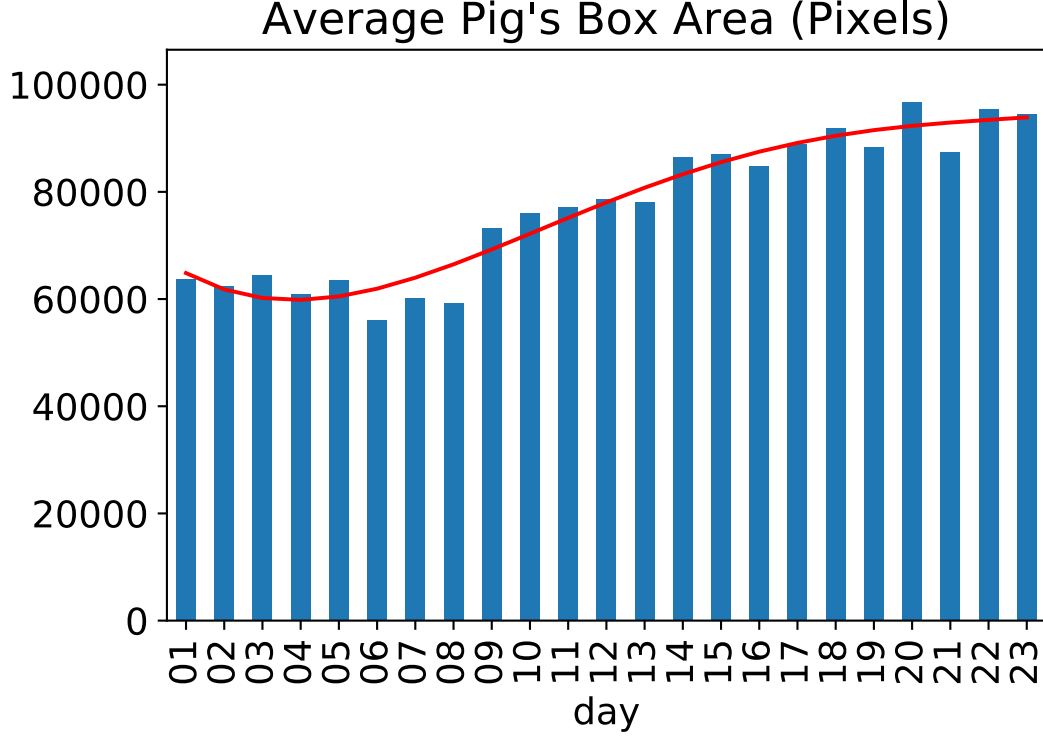


Figure 3: Changes of the estimated bounding box mean area on the full dataset through the acquisition days.

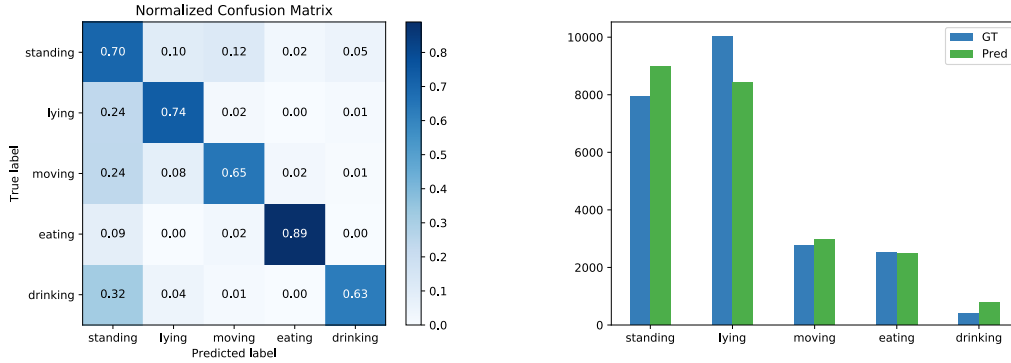


Figure 4: Confusion matrix for the 5 behaviors on the validation set (left). Distributions of the GT and predicted behaviors on the validation set (right)

together comparing the Intersection over Union and their appearance and finding the best assignments with the Kuhn-Munkres algorithm (Kuhn, 1955). If a detection is not matched to any tracker, a new one is initialized while, if a tracker is not matched to a detection for 8 frames, the tracker is removed. As tracker, we employ the MOSSE (Bolme et al., 2010) algorithm.

We evaluate the quality of the tracking algorithm using the following metrics:

- Multiple Object Tracking Accuracy (*MOTA*) (Bernardin and Stiefelhagen, 2008; Milan et al., 2016) combining three sources of errors as:

$$MOTA = 1 - \frac{\sum_t (FN_t + FP_t + IDSW_t)}{\sum_t GT_t} \quad (1)$$

where FN is a tracker hypothesis which is wrongly not generated, FP is a tracker hypothesis generated where there is none in the ground truth, $IDSW$ is a mismatch between the current and previous association and GT is the number of ground truth objects;

- Identification F_1 (IDF_1) score (Ristani et al., 2016) representing the ratio of correctly identified detections over the average number of ground truth and computed detections:

$$IDF_1 = \frac{2IDTP}{2IDTP + IDFP + IDFN} \quad (2)$$

which differs from the $MOTA$ as it performs a 1-to-1 mapping between IDs, without considering identity switches or fragmentations of the predicted trajectories;

- Number of identity switches, occurring when the tracker jumps from one identity to another;
- Number of fragmentations, accounting for tracklet switches between missed and not missed states.
- Average tracklet length, which summarizes the tracker effectiveness in following the pigs through the sequence (a perfect result would be 8 tracks, each with 1800 frames).

Table 3 reports the results on the validation set. While there is some variance between sequences, most of the pigs are tracked for long periods, they are rarely swapped and few false positives occur. In particular, the average tracklet length is more than half a sequence (i.e. more than 2.5 minutes) and the per-sequence number of switches between two pigs is only 11 (i.e. on average, each pig track switches about 1.5 times).

4.3 Behavior Analysis

Behavior analysis uses the detections and tracklets identified by the algorithms from Sec 4.1 and Sec 4.2 to predict a behavior class for each pig in every frame. While it is possible to directly predict the behavior along with the pig detection, a single-frame approach like the one employed in Sec 4.1 would struggle to correctly identify behaviors that depend on multiple frames, such as moving or standing. Here, a combination of deep learning based and traditional techniques is used to better fit the different natures of the behaviors of interest.

The first step computes the average movement of the pig, as the movement of bounding box centroid locations in a given time-frame. The average depth inside the bounding box is used to predict the average pig movement in centimeters and compared to a fixed threshold in the same unit. In this way a single threshold can be applied to pigs in any part of the pen. We use a threshold of 2.5 cm over the center of mass movement in a 2 seconds window and show that it is enough to sufficiently discriminate between unmoving and moving behaviors.

When the initial decision yields unmoving, we identify whether a pig is feeding or drinking by its distance and orientation from the feeder or the drinkers. Because we have collected our data from a single pen, the positions of those items is known. However, because the annotated bounding boxes do not hold any orientation information, identifying the pig orientation is not trivial. In practice, we compute the gray-scale image moments (Ming-Kuei Hu, 1962) on each bounding box and extract the pig center of mass and angle from a combination of the first and second order central moments, under the hypothesis of having ellipse-shaped entities, which is a good assumption for pigs (Brünger et al., 2020). It is worth noting how this approach cannot disambiguate between 2 angles spanned by π like a pig facing or giving its back to the feeder. In practice, we notice the latter happens very rarely, as other pigs are likely to step in to feed frequently.

The remaining behaviors consist of lying and standing. These actions do not depend on specific locations in the pen, and the appearance of the pig must be taken into consideration for choosing between them. Our first approach made use of the depth information but proved unreliable (see Sec 3). Therefore, a deep-learning method based on ResNet18 (He et al., 2016) is used to classify the bounding box into one of the classes of interest. The network is trained on the training split from Sec 4.1 and validated on the validation split. We compensate for class imbalance by inverse weighting during training (i.e. samples from the most common classes are weighted less than samples from the uncommon classes).

We report results in terms of accuracy on the validation set for the five behaviors in Fig. 4 (left). It is likely that more sophisticated, generic, and accurate behavior classification methods exist, but we reiterate one claim of the paper: the collective behavior statistics are accurate, even though individual frame-level labels may not always be as accurate (about 73% accurate on average over the unbalanced validation set, of which about 75% of the frames were either standing or lying). To support this claim, we report in Fig. 4 (right) two distribution histograms (for the ground-truth and the predicted behaviors, again on the validation set). It can be observed that these two distributions are very similar, with a KL divergence value of 0.014. As another measure of quality, we compute the average global prediction error $\sum_i |GT_i - Pred_i| / \sum_i GT_i = 0.14$, which shows that the individual errors tend to cancel out to give more accurate collective statistics.

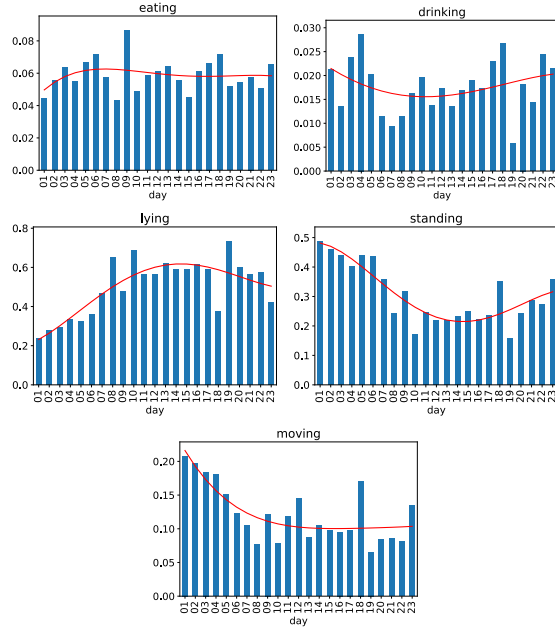


Figure 5: Temporal graphs of the behavior changes aggregated by day on the full dataset. The red solid line shows an interpolation of the data

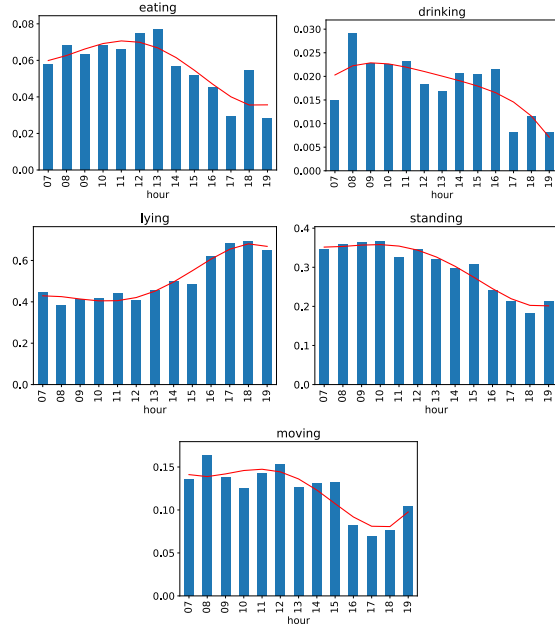


Figure 6: Temporal graphs of the behavior changes aggregated by hour on the full dataset. The red solid line shows an interpolation of the data

5 DISCUSSION

We report in this section our observations on the full unlabeled dataset, after applying our pipeline for behavior understanding. We aggregate the results for the predicted behaviors by days (Fig. 5) and by daytime hours (Fig. 6). The statistics are computed over approximately 27 million pig detections, which means approximately 1 million detections per day in Fig. 5 and 2 million detections per time of day in Fig. 6. From the former we draw the following conclusions:

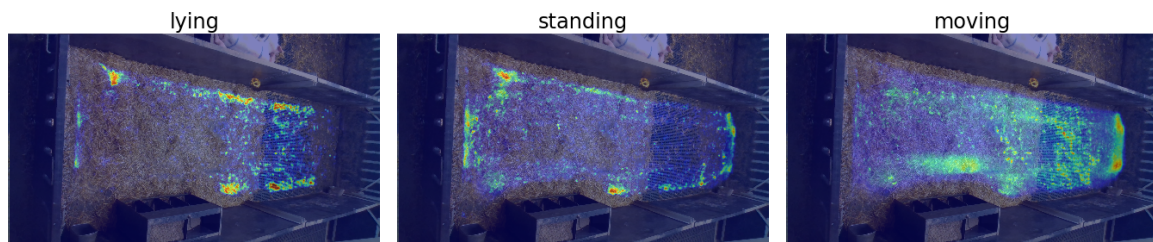


Figure 7: Heatmaps for 3 out of the 5 behaviors, computed from the bounding boxes detected on the full dataset. We omit eating and drinking as those behaviors are directly identified using the bounding box location. Best viewed in color.

- **Eating** and **drinking** behaviors do not vary drastically during the observation period. This matches the ad libitum availability of food and water that the animals were provided with. These two actions are performed for a total of around 10% of the whole time. For drinking, the indoor ventilated setting reduces the need of water;
- **Moving**, **standing** and **lying** follow a mirrored pattern. While the first two decrease through time, the latter drastically increases. This matches the expected behavioral pattern of growing pigs in a new environment. The first days are characterized by high levels of activity. This is due to various factors, including the pigs' youth, being in a new environment, the presence of other pigs and not being used to daily inspections among others. After these first days, they rapidly adapt to the new situation while at the same time they begin to grow more quickly (see Fig. 3). This eventually results in pigs spending most of the time lying and/or sleeping.

On the other hand, the analysis over the daytime highlights how pigs in these conditions (indoor, artificial light only) are mainly diurnal, where activities (moving, eating and drinking) are performed intensively during the morning and early afternoon and the animals are less active during the late afternoon.

We also visualize the heatmaps for 3 of the 5 behaviors in Fig. 7. These are computed by plotting the centroid of the detected bounding boxes for a single behavior over the 6 weeks period. It is worth noting:

- **Lying** rarely occurs in front of the feeder. This is because pigs keep alternating to eat, making the area crowded. The areas where lying occurs more often are in fact those along the edges, but not at the very far right end where much toilet behavior occurs due to the slatted floor.
- **Standing** is more spread around, with a preference for the left part of the pen;
- **Moving** is focused in two areas mainly. The first one is the right section of the pen. This is where pigs usually run when operators move along the aisle near the left edge of the pen and it's also the area deemed as toilet. The second area is in front of the feeder, as pigs move here to access and leave the feeder itself.

6 CONCLUSIONS

We presented here a detection-tracking-behavior pipeline for long-term behavior changes of individual pigs in an indoor pen. This analysis is powered by our new large pig dataset which includes annotations for various tasks, and which we will publicly release with no restrictions. The conclusions drawn from the aggregated data match the expectations of experts, and justify our claim that collective behavior statistics are accurate, even though individual frame-level labels may not always be as accurate. This is valid not only for actions performed frequently (e.g. lying), but also for those occurring less often (e.g. eating or drinking).

Future improvements can be envisioned for this challenging task. On the one hand, single components (e.g. the detection algorithm) could be specialized for the setting. On the other hand, given that the errors in the different stages of the pipeline compound, a single end-to-end method for detection-tracking-behavior is also a possible future outcome. The detection ground-truth could be refined to use ellipses instead of axis-aligned bounding boxes. Another direction for extensions is increasing the number or breakdown of the behavior classifications.

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